



4



8



4



8



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A								
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D								
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Sheet 1 of 17
Classification: ITAR Restricted
Space Systems Engineering
Direct RF Sampling Acquisition System

Sheet: /Title/
File: Sheet01_Title.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3

Date: 2026-09-10

Rev: 1.0

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 2/18

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SMPA Input, Balun, AC Coupling
Sheet 2: RF Input Section

Sheet: /RF_Input/
File: Sheet02_RF_Input.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/18

	1	2	3	4	5	6	7	8
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D								
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12-bit 5.2GSPS ADC with JESD204C
Sheet 3: ADC12DJ5200-SP Configuration

Sheet: /ADC/
File: Sheet03_ADC.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 4/18	

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D								
E								
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Space-Grade Radiation Tolerant FPGA
Sheet 4: FPGA XQRVC1902 Configuration

Sheet: /FPGA/
File: Sheet04_FPGA.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 5/18	

	1	2	3	4	5	6	7	8
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LMX2615-SP VCO, LMK04832-SP Jitter Cleaner, VCXO
Sheet 5: Clock Tree

Sheet: /Clock/
File: Sheet05_Clock.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3Date: 2026-09-10

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: 1.0
Id: 6/18

	1	2	3	4	5	6	7	8
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B								
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F	1	2	3	4	5	6	7	8

4.5V Input, TVS Protection, EMI Filter, Reverse Polarity
Sheet 6: Power Input Section

Sheet: /Power_Input/
File: Sheet06_PowerInput.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/18

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B								
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E								
F	1	2	3	4	5	6	7	8

VCCINT_1V0 Rail for FPGA Core
Sheet 7: TPS7H5002-SP, 44A Output, 100nH Inductor

Sheet: /Buck_1V0_Core/
File: Sheet07_Buck_1V0.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/18

	1	2	3	4	5	6	7	8
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F	1	2	3	4	5	6	7	8

VDDIO_1V8 Rail
Sheet 8: ISL70003ASEH, 6A Output, 2.2uH Inductor

Sheet: /Buck_1V8_Aux/
File: Sheet08_Buck_1V8.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/18

	1	2	3	4	5	6	7	8
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E								
F	1	2	3	4	5	6	7	8

VDDIO_3V3 Rail
Sheet 9: ISL70003ASEH, 3A Output, 4.7uH Inductor

Sheet: /Buck_3V3_I0/
File: Sheet09_Buck_3V3.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3

Date: 2026-09-10

Rev: 1.0

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 10/18

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VANA_1V0 Rail
Sheet 10: TPS7H1111-SP, 1A, Low Noise for ADC AVDD

Sheet: /LDO_1V0_Analog/
File: Sheet10_LDO_1V0.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 11/18	

	1	2	3	4	5	6	7	8
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B								
C								
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E								
F	1	2	3	4	5	6	7	8

VDIG_1V8 Rail
Sheet 11: TPS7H1111-SP, 1A, Digital DVDD

Sheet: /LDO_1V8_Digital/
File: Sheet11_LDO_1V8.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 12/18

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B								
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D								
E								
F	1	2	3	4	5	6	7	8

VCLK_2V5 Rail
Sheet 12: TPS7H1121–SP, 300mA, Ultra Low Noise for Clock

Sheet: /LDO_2V5_Clock/
File: Sheet12_LDO_2V5.kicad_sch

Title: X–Band Digital Module 20x20cm

Size: A3

Date: 2026–09–10

Rev: 1.0

KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4

Id: 13/18

	1	2	3	4	5	6	7	8
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E								
F	1	2	3	4	5	6	7	8

Sequencing: VCCINT -> VDDIO -> VANA -> VDIG -> VCLK
Sheet 13: 2x TPS7H3014-SP, 5-Rail Supervision + Daisy-Chain

Sheet: /Sequencer/
File: Sheet13_Sequencer.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 14/18	

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B								
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D								
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Sheet 14: JTAG, SpaceWire, Power, UART, GPIO		
Sheet: /Connectors/ File: Sheet14_Connectors.kicad_sch		
Title: X-Band Digital Module 20x20cm		
Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 15/18

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E								
F	1	2	3	4	5	6	7	8

Sheet 15: All Rails and Signals		
Sheet: /TestPoints/ File: Sheet15_TestPoints.kicad_sch		
Title: X-Band Digital Module 20x20cm		
Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 16/18	

	1	2	3	4	5	6	7	8
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B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Sheet 16: Bypass Capacitors per IC Group		
Sheet: /Decoupling/ File: Sheet16_Decoupling.kicad_sch		
Title: X-Band Digital Module 20x20cm		
Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 17/18

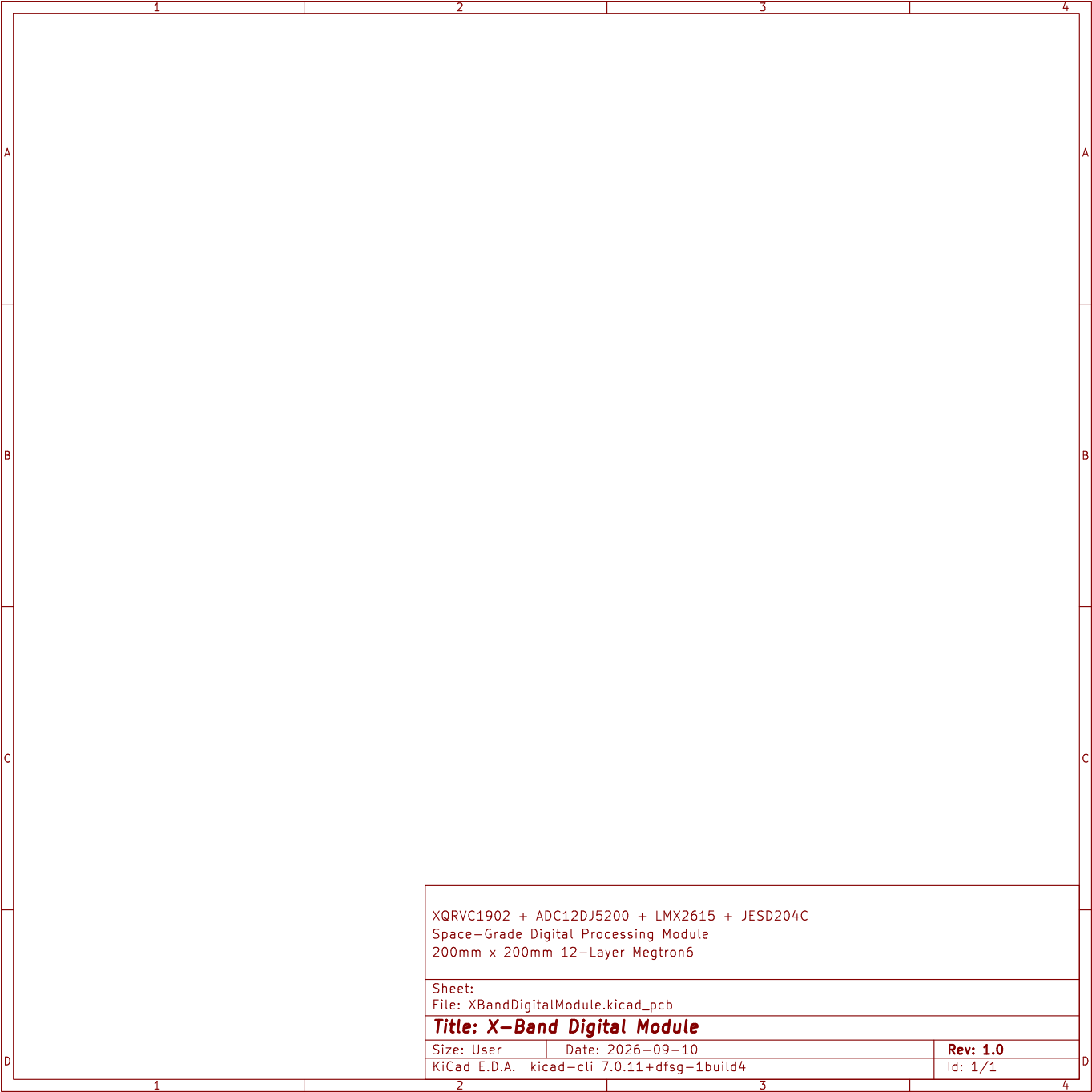
	1	2	3	4	5	6	7	8
A								
B								
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D								
E								
F								

VIN 4.5V -> 6 rails via 3 buck + 3 LDO
Sheet 17: Power Block Diagram and Sequencing

Sheet: /PowerTree/
File: Sheet17_PowerTree.kicad_sch

Title: X-Band Digital Module 20x20cm

Size: A3	Date: 2026-09-10	Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 18/18	



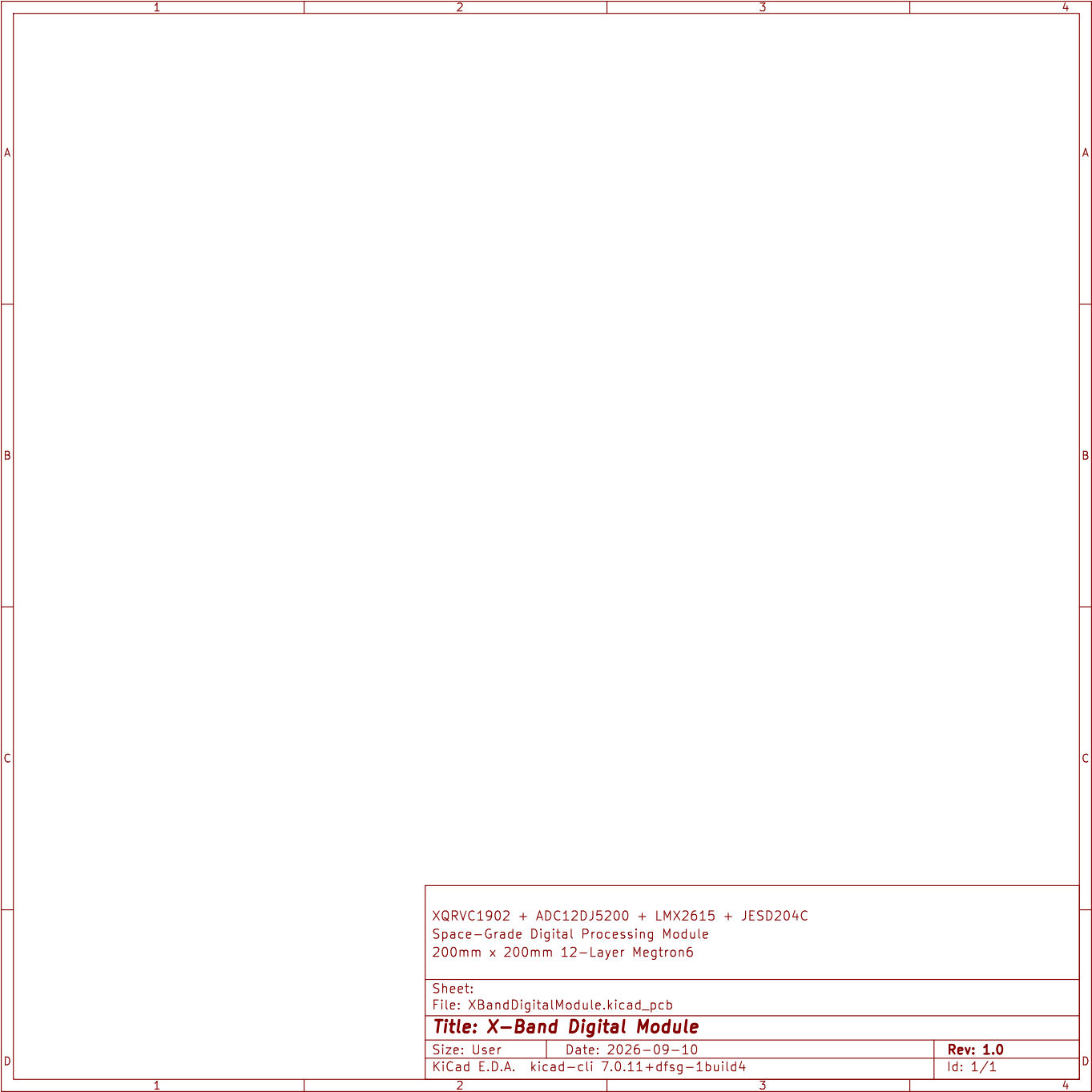
XQRVC1902 + ADC12DJ5200 + LMX2615 + JESD204C
Space-Grade Digital Processing Module
200mm x 200mm 12-Layer Megtron6

Sheet:
File: XBandDigitalModule.kicad_pcb

Title: X-Band Digital Module

Size: User Date: 2026-09-10
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: 1.0
Id: 1/1



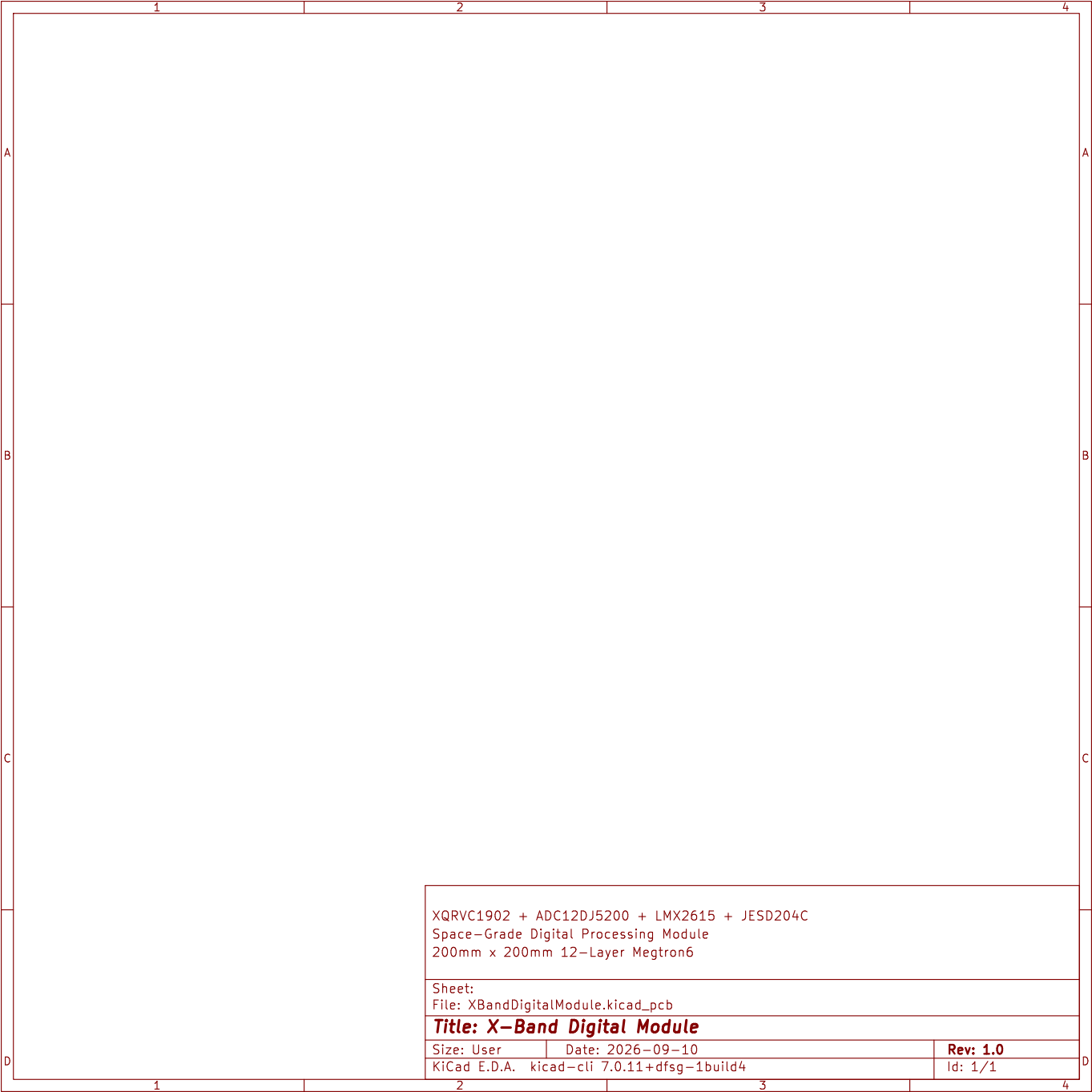
XQRVC1902 + ADC12DJ5200 + LMX2615 + JESD204C
Space-Grade Digital Processing Module
200mm x 200mm 12-Layer Megtron6

Sheet:
File: XBandDigitalModule.kicad_pcb

Title: X-Band Digital Module

Size: User Date: 2026-09-10
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: 1.0
Id: 1/1



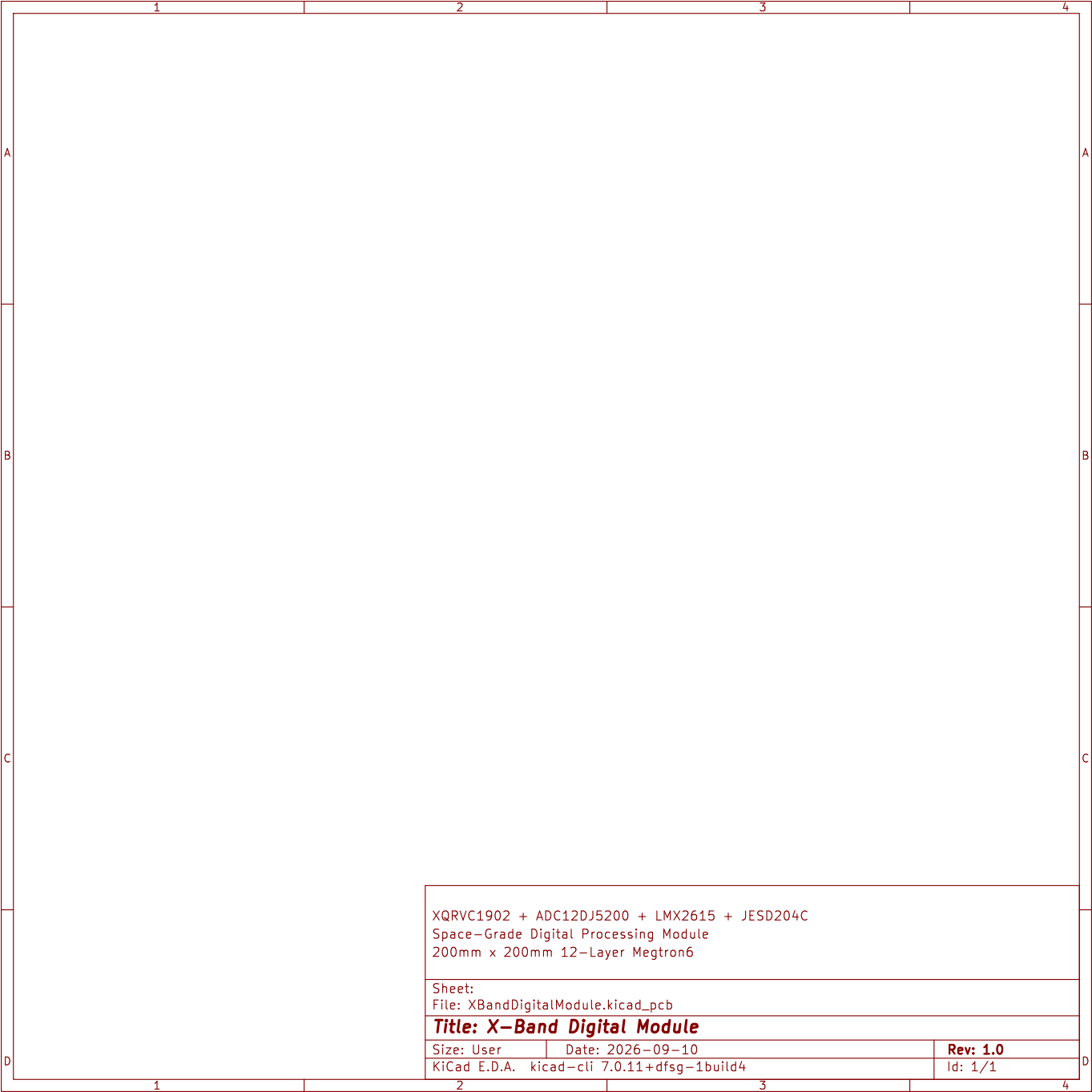
XQRVC1902 + ADC12DJ5200 + LMX2615 + JESD204C
Space-Grade Digital Processing Module
200mm x 200mm 12-Layer Megtron6

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File: XBandDigitalModule.kicad_pcb

Title: X-Band Digital Module

Size: User Date: 2026-09-10
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: 1.0
Id: 1/1



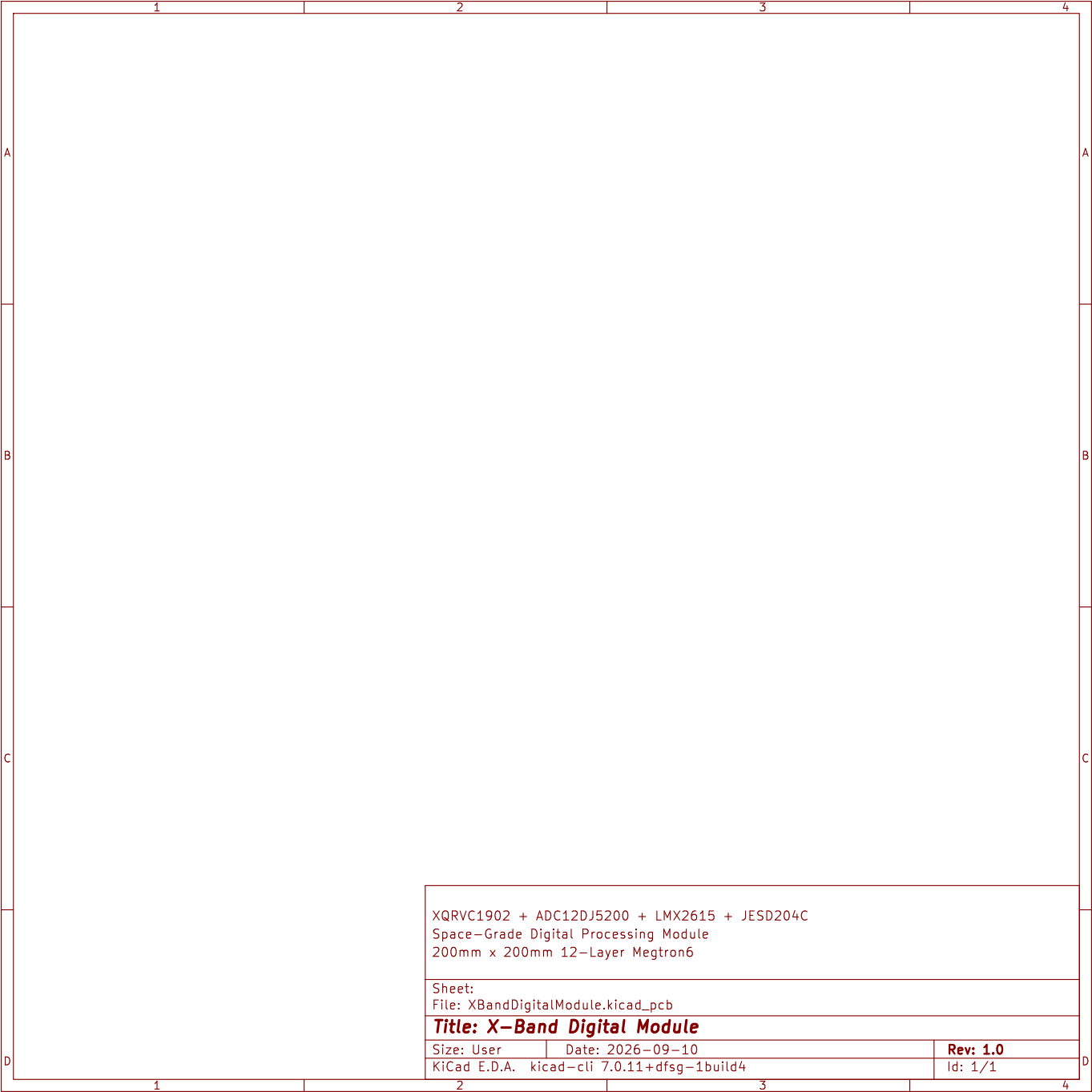
XQRVC1902 + ADC12DJ5200 + LMX2615 + JESD204C
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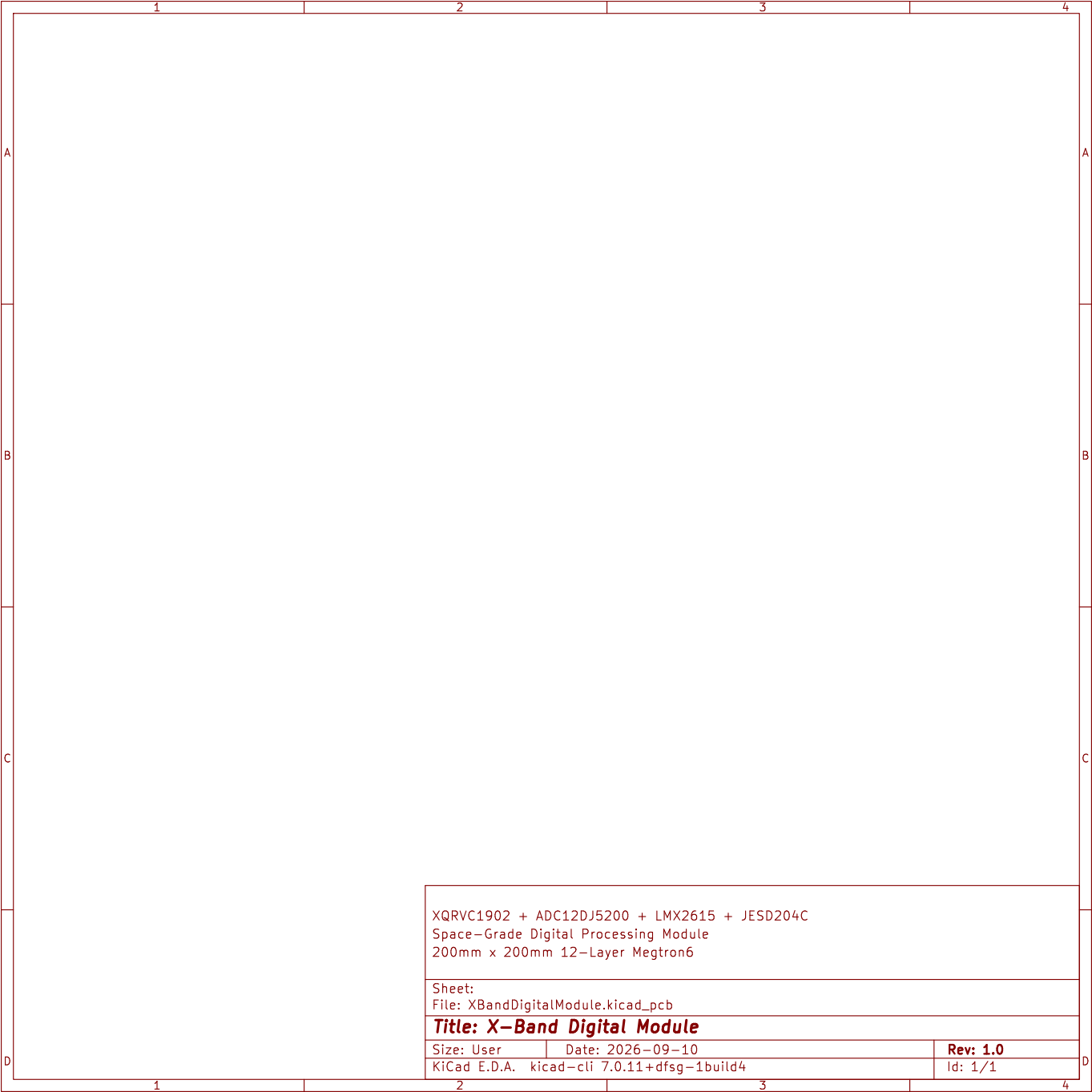
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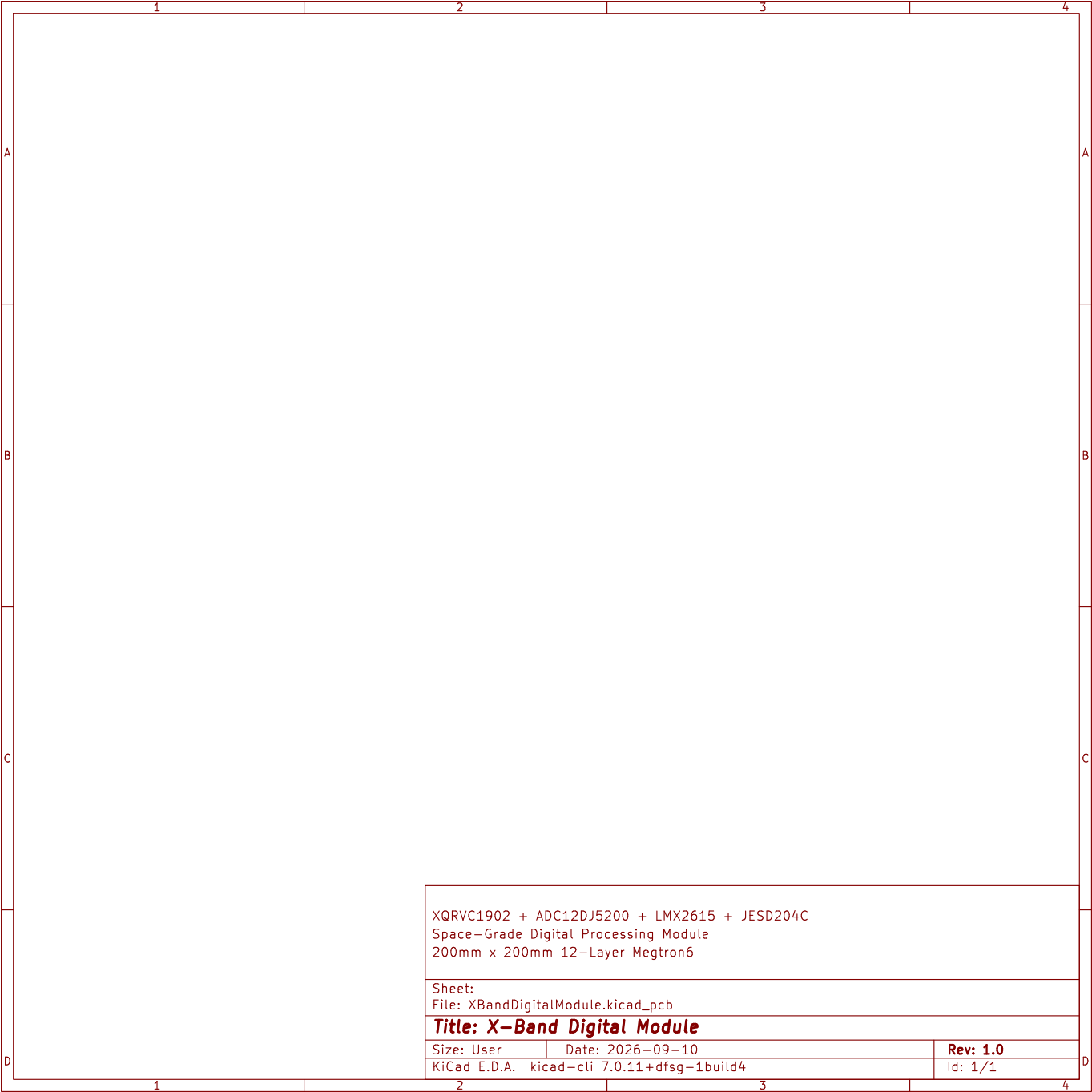
Title: X-Band Digital Module

Size: User Date: 2026-09-10
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: 1.0
Id: 1/1







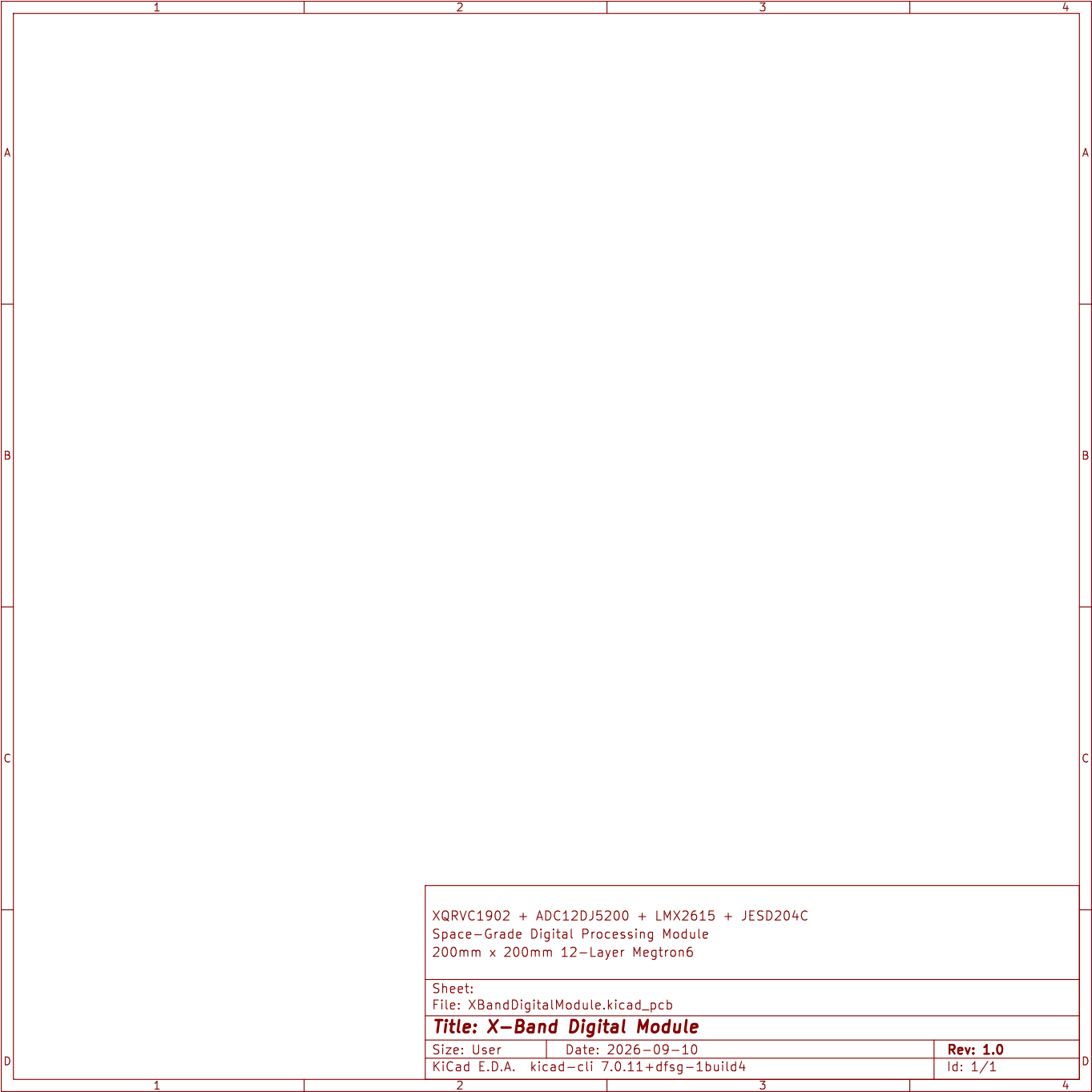
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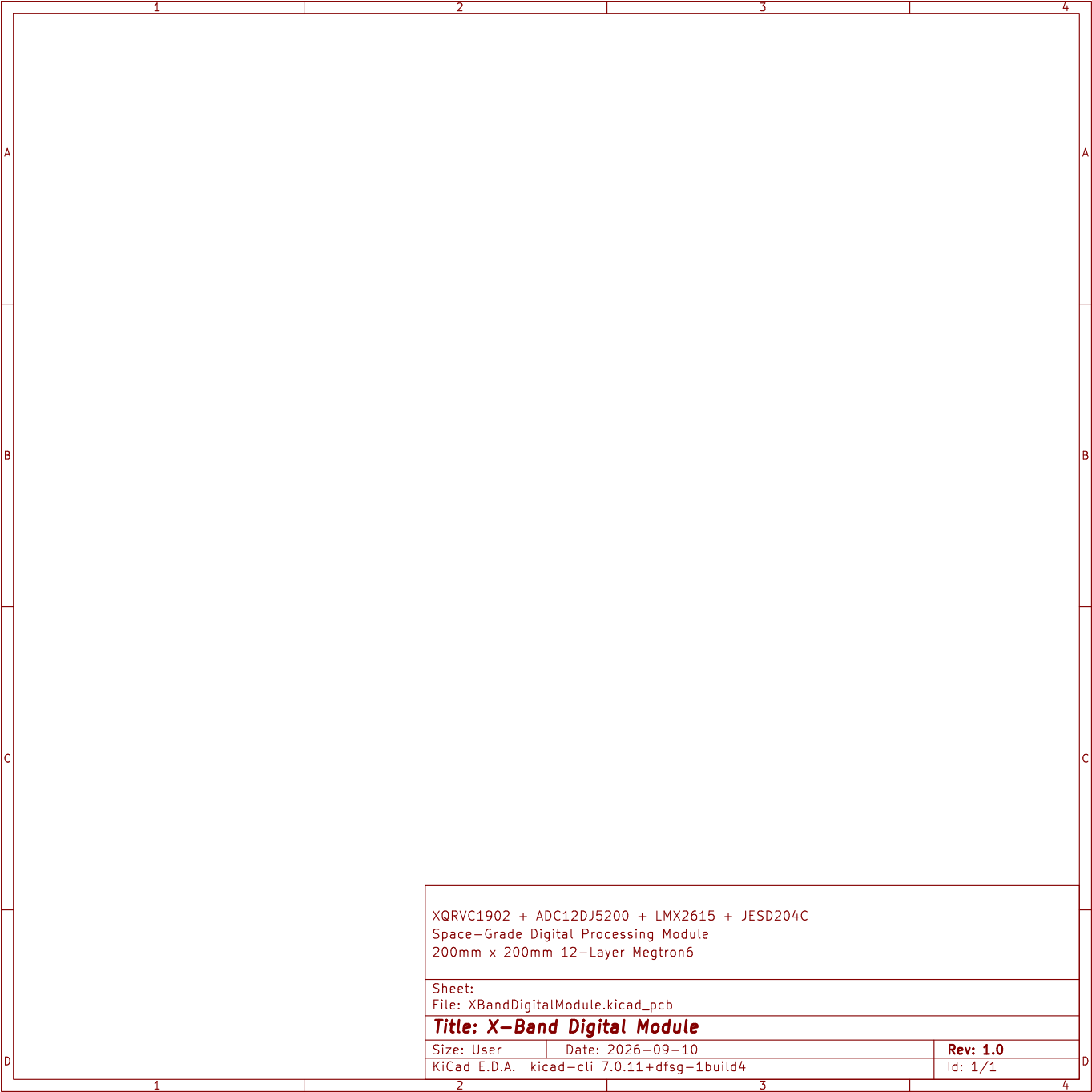


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Space-Grade Digital Processing Module
200mm x 200mm 12-Layer Megtron6

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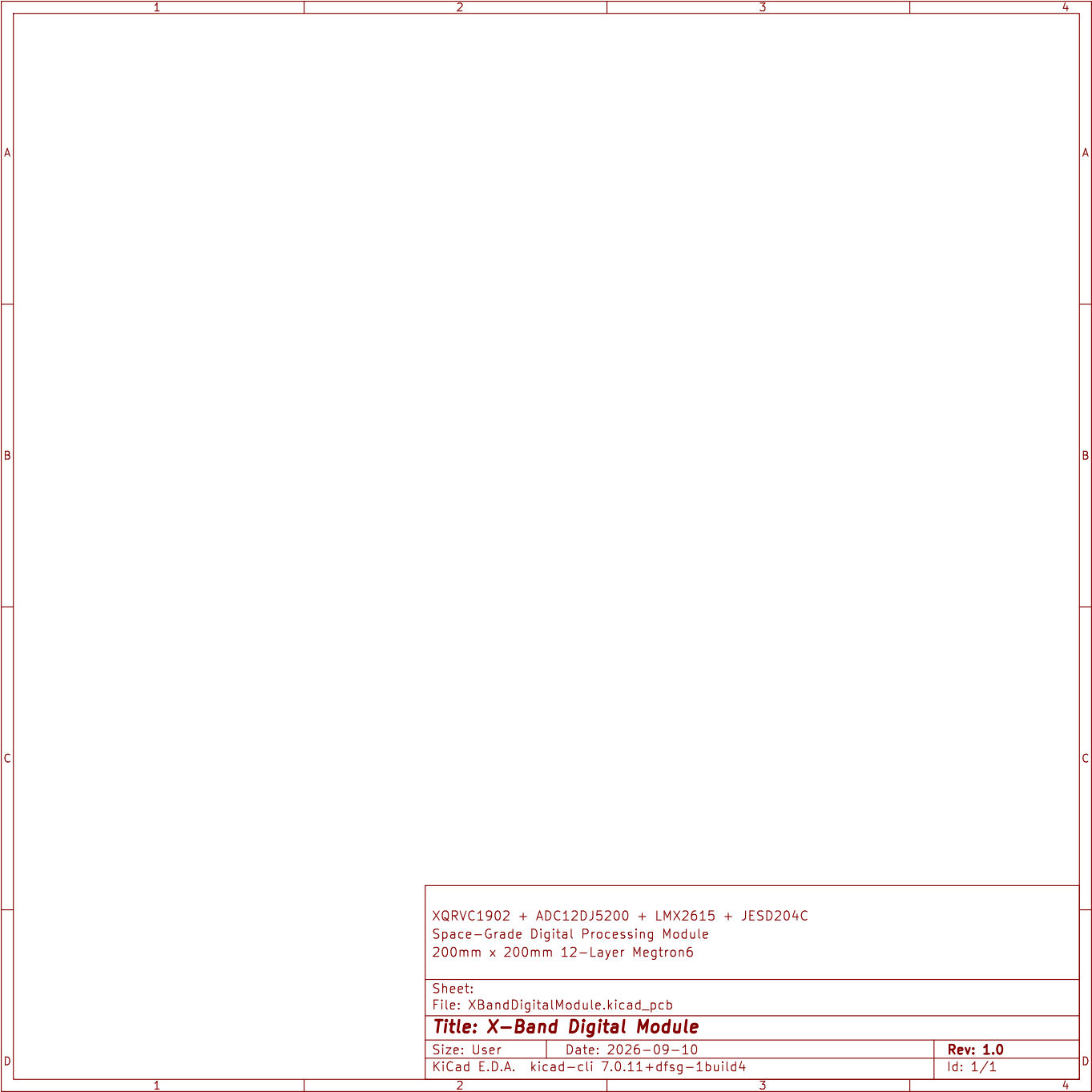


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Space-Grade Digital Processing Module
200mm x 200mm 12-Layer Megtron6

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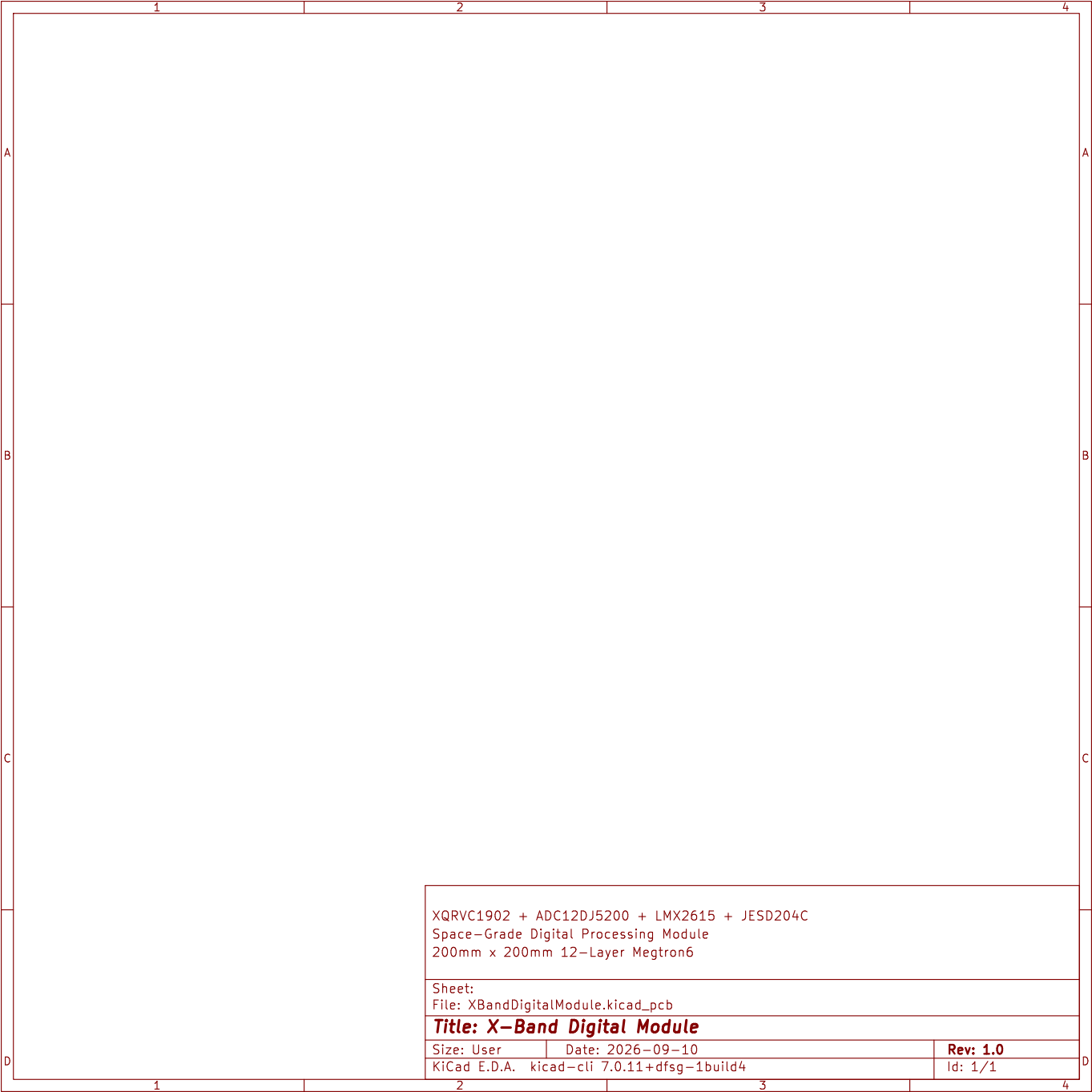
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Rev: 1.0
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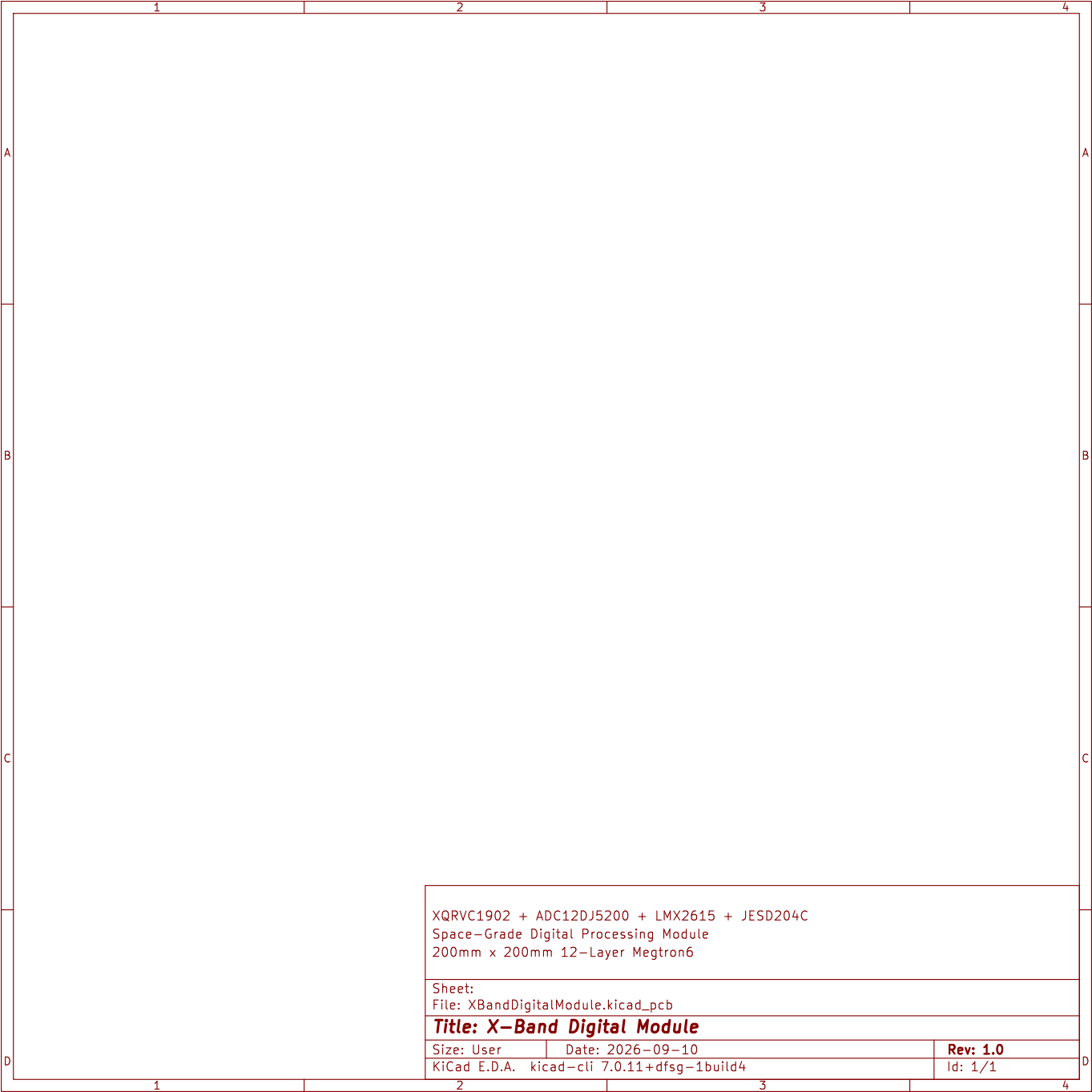
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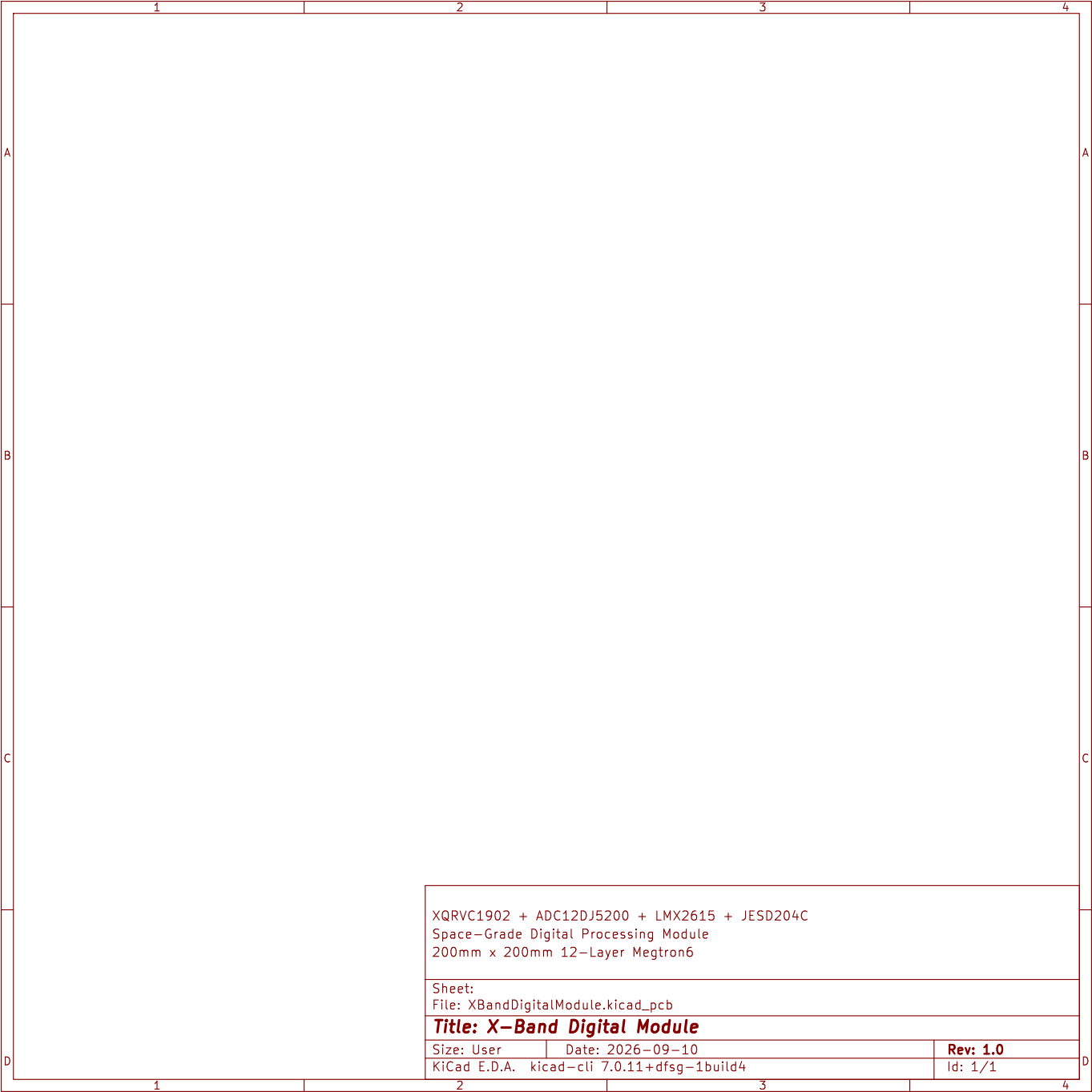
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